



Did ChatGPT Kill Stack Overflow?

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Programmers did not stop asking questions. They started asking machines.

Abstract

For more than a decade, Stack Overflow was one of the internet's central programming knowledge systems. This short exploratory analysis examines whether the rise of ChatGPT and AI-assisted programming tools coincided with declining public activity on Stack Overflow. Using Stack Exchange activity data, Stack Overflow Developer Survey statistics, and Google Trends data, the analysis finds a sharp decline in monthly questions after late 2022, especially across beginner-oriented programming tags. The evidence does not prove direct causality, but it strongly suggests that AI-assisted programming changed how developers search for technical answers online.

Keywords: ChatGPT; Stack Overflow; AI coding assistants; Google Trends; developer behaviour; data analysis

Repository: https://github.com/Alfemil199/did_chatgpt_kill_stackoverflow

The Shift

For more than a decade, Stack Overflow was one of the internet's central programming knowledge systems. Developers searched for syntax help, debugging assistance, framework explanations, and implementation examples through public question-and-answer threads. But large language models changed the interface itself. Instead of searching through archived discussions, developers could suddenly ask conversational systems directly and receive contextual answers instantly. This report asks a deliberately provocative question:

Did ChatGPT kill Stack Overflow?

This report does not attempt to prove direct causality. Instead, it examines whether several public data signals changed in the same period as conversational AI tools became mainstream.

Data and Method

The analysis uses three lightweight public data sources.

Primary source: Stack Overflow activity

Monthly Stack Overflow activity was extracted from Stack Exchange Data Explorer [1]. The exported datasets contain monthly question counts and tag-specific monthly question counts for Python, JavaScript, HTML, CSS, and Java. This is the primary evidence because it directly measures public activity on Stack Overflow.

Context source: Stack Overflow Developer Survey

Selected AI-related statistics from the Stack Overflow Developer Surveys 2023–2025 were used to contextualise developer adoption of AI-assisted programming tools [2][3][4]. This source is treated cautiously because the survey is distributed through Stack Overflow's own ecosystem and may overrepresent developers who are already familiar with the platform.

Attention source: Google Trends

Google Trends was used as a broad attention signal for ChatGPT [5]. Google Trends measures relative search interest, not absolute usage. It is therefore used only as supporting context, not as direct evidence of developer migration.

Stack Overflow Activity Collapsed

The first signal is the overall number of questions posted on Stack Overflow. A gradual decline is visible before late 2022, but the drop becomes much sharper after the public release of ChatGPT.

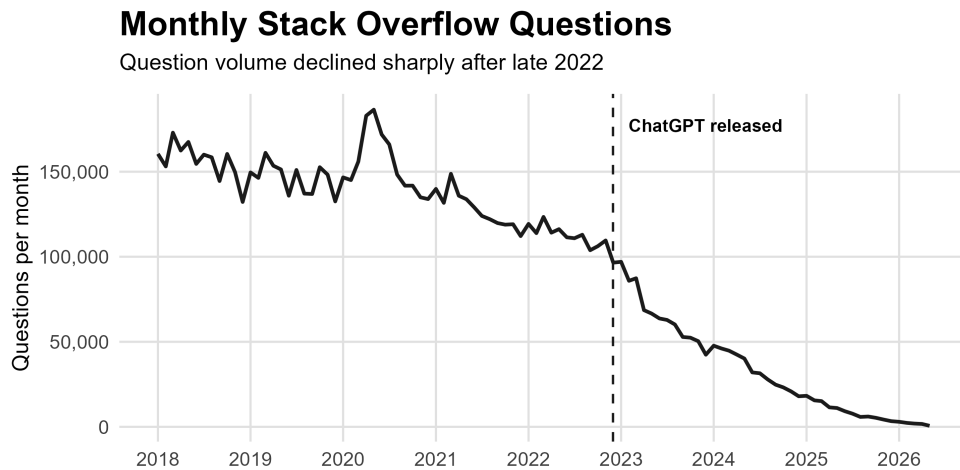


Figure 1. Monthly Stack Overflow question volume, 2018–2026.

Beginner Workflows Declined Together

The strongest signal is not only that Stack Overflow declined overall. It is that common beginner-oriented programming tags declined together. Python, JavaScript, HTML, CSS, and Java all show strong declines after late 2022. These are precisely the kinds of repetitive syntax, debugging, and introductory coding questions that modern LLM-based assistants can often answer directly.

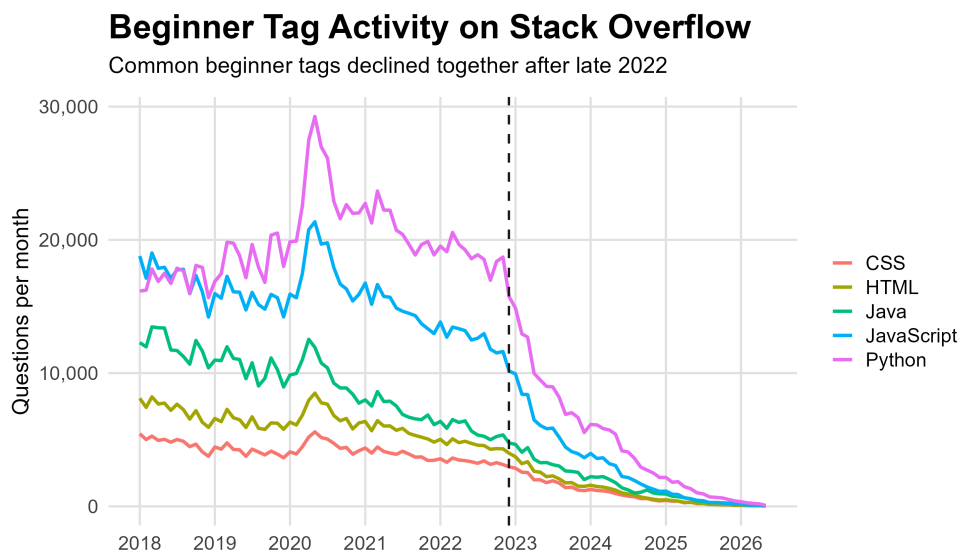


Figure 2. Monthly activity for common beginner-oriented Stack Overflow tags.

The Post-ChatGPT Average Is Much Lower

To summarise the shift, monthly activity was grouped into two periods: before and after the public release of ChatGPT. Average monthly Stack Overflow questions decreased from approximately 140,000 questions per month before ChatGPT to roughly 33,000 questions per month afterwards.

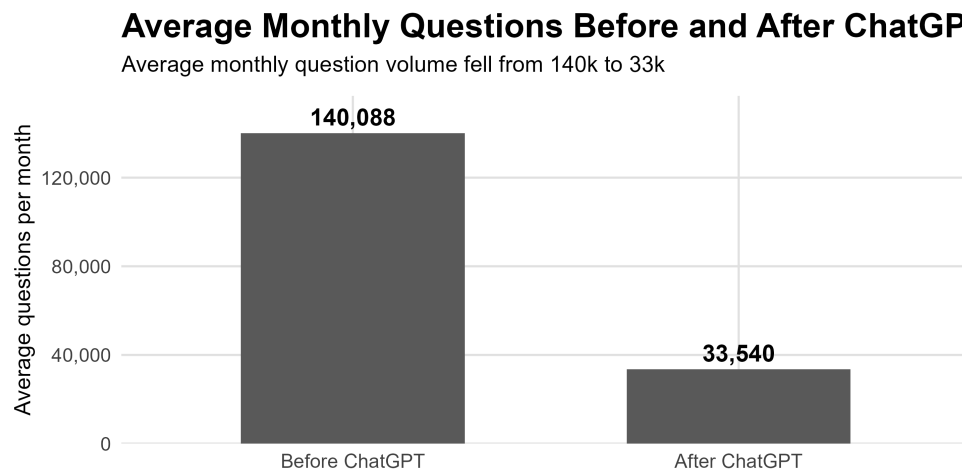


Figure 3. Average monthly Stack Overflow questions before and after the release of ChatGPT.

Developers Rapidly Adopted AI

At the same time that Stack Overflow activity declined, developer adoption of AI-assisted programming tools increased rapidly. According to Stack Overflow Developer Surveys between 2023 and 2025, the share of developers using or planning to use AI tools rose from 70% to 84%.

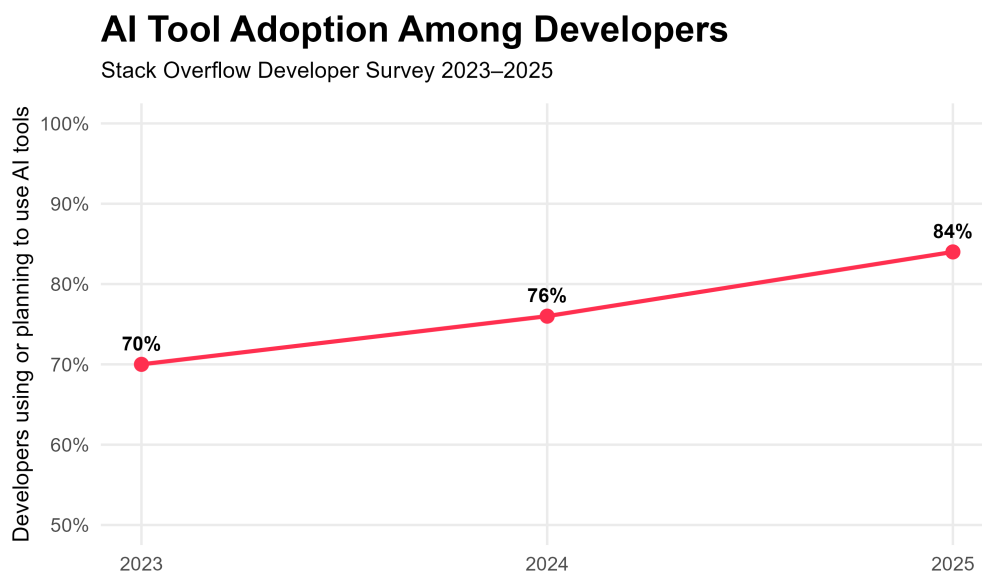


Figure 4. AI tool adoption among developers according to Stack Overflow Developer Surveys, 2023–2025.

Public Attention Shifted Toward ChatGPT

Google Trends data shows a rapid increase in global search interest for “ChatGPT” after late 2022.

This figure is not used as proof that developers replaced Stack Overflow with ChatGPT. Searches for ChatGPT include many non-programming use cases. Instead, it is used as a broad public-attention signal showing how quickly conversational AI entered mainstream awareness.

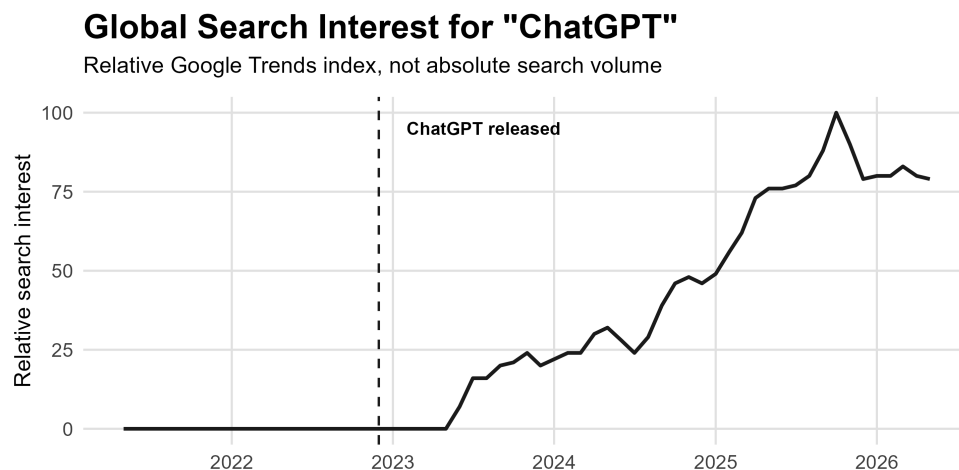


Figure 5. Global Google Trends search interest for “ChatGPT”. The index is relative, not absolute search volume.

Interpretation

The data suggests a strong overlap between the rise of conversational AI systems and declining Stack Overflow activity. The most defensible interpretation is not that ChatGPT literally killed Stack Overflow. The stronger claim is that AI-assisted programming appears to have accelerated a shift in how developers access programming knowledge.

Stack Overflow was built around searchable public troubleshooting. Large language models changed the interface. Developers no longer need to search through dozens of archived answers when they can ask a conversational model for a contextual explanation, a code example, or a debugging suggestion.

This does not make Stack Overflow irrelevant. It may remain valuable as a technical archive, source of high-quality historical answers, and training substrate for broader programming knowledge. But its role as the default first stop for everyday programming help appears to be shrinking.

Technical Workflow

The analysis was conducted using a lightweight reproducible workflow built primarily in R.

- Stack Overflow activity data was collected from Stack Exchange Data Explorer.
- Google Trends data was exported directly from Google Trends.
- Stack Overflow Developer Survey statistics were manually extracted from the official annual survey reports.

- Data cleaning, aggregation, and visualization were performed in R using packages from the tidyverse ecosystem, primarily dplyr, ggplot2, and readr.
- Figures were exported from R and later adapted into editorial-style infographic variants for social media distribution and visual communication.
- The final article and layout were written in LaTeX using a custom document class developed for independent research publications.

Limitations

This analysis is exploratory and does not establish direct causality. Several alternative explanations may also contribute to the decline:

- search engine changes,
- changing developer culture,
- better official documentation,
- more use of Discord, Reddit, GitHub Issues, and YouTube,
- stricter moderation or duplicate-question closure patterns,
- declining participation in public Q&A communities.

Google Trends is only an attention proxy. Stack Overflow survey data may also reflect the platform's own user base rather than all developers.

The key claim is therefore not causal proof. The claim is that several independent public signals changed in the same period as AI-assisted programming tools became mainstream.

Conclusion

So, did ChatGPT kill Stack Overflow? Not literally.

But the available data suggests that AI-assisted programming tools coincided with a substantial decline in Stack Overflow activity between 2022 and 2026. The clearest signal is not only that total question volume declined. It is that beginner-oriented programming tags declined together, while developers increasingly adopted AI tools and public attention shifted sharply toward ChatGPT. Stack Overflow remains an important technical archive. But the public troubleshooting workflow that defined the platform for more than a decade appears to be shrinking rapidly in the age of AI-assisted programming.

The stronger conclusion is simple: AI-assisted programming changed how developers search for answers online.

Data and References

1. Stack Exchange Data Explorer:
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2. Stack Overflow Developer Survey 2023:
<https://survey.stackoverflow.co/2023#ai>
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<https://survey.stackoverflow.co/2024/ai/>
4. Stack Overflow Developer Survey 2025:
<https://survey.stackoverflow.co/2025/ai#sentiment-and-usage-ai-select-ai-select>
5. Google Trends:
<https://trends.google.com/trends/>